the vertebral column and ventrally to the sternum. It has two articulation surfaces on its dorsal end, hence called bicephalic. The first seven pairs of ribs are called '**true ribs**' or **vertebro–sternal ribs**. Dorsally they are attached to the thoracic vertebrae and ventrally connected to the sternum with the help of hyaline cartilages. The 8th, 9th and 10th pairs of ribs do not articulate directly with the sternum but joined with the cartilaginous (hyaline cartilage) part of the seventh rib. These are called **'false ribs'** or **vertebro-chondral ribs**. The last 11th and 12th pairs of ribs are not connected ventrally. Therefore, they are called as **'floating ribs'** or **vertebral ribs**. Thoracic vertebrae, ribs and sternum together form the ribcage.

Rib cage protects the lungs, heart, liver and also plays a role in breathing.

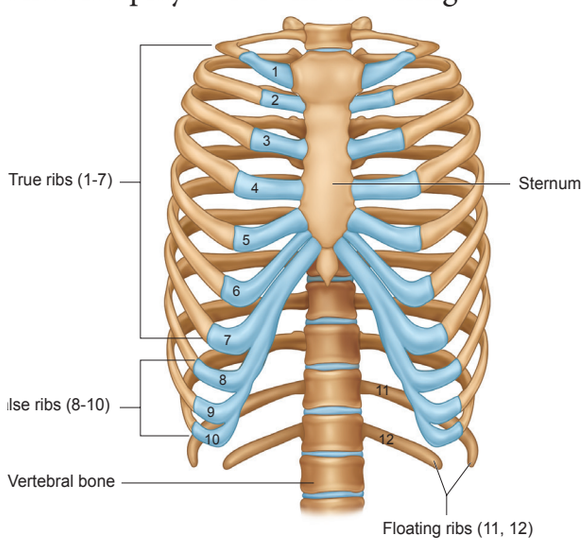


Figure 9.7 Rib cage

9.9 The Appendicular skeleton

The bones of the upper and lower limbs along with their girdles constitute the appendicular skeleton. The appendicular skeleton is composed of 126 bones.

(a) The Pectoral girdle

The upper limbs are attached to the pectoral girdles. These are very light and allow the upper limbs a degree of mobility not seen anywhere else in the body. The girdle is formed of two halves**.** Each half of the pectoral girdle (Figure 9.8) consists of **a clavicle** or **collar bone** and a **scapula**. The scapula is a large, thin, triangular bone situated in the dorsal surface of the ribcage between the second and seventh ribs. It has a slightly elevated ridge called the spine which projects as a flat, expanded process called the **acromion**. The clavicle articulates with this process. Below the acromion is a depression called the **glenoid cavity** which articulates with the head of the humerus to form the shoulder joint. Each clavicle is a long slender bone with two curvatures which lies horizontally and connects axial skeleton with appendicular skeleton.

The Upper limb

The upper limb consists of 30 separate bones and is specialized for mobility. The skeleton of the arm, the region between the shoulder and elbow is the **humerus**. The head of humerus articulates with the **glenoid cavity** of the scapula and forms the shoulder joint. The distal end of humerus articulates with the two forearm bones the **radius** and **ulna**. The forearm is the region between the elbow and the wrist. **Olecranon process** is situated at the upper end of the ulna which forms the pointed portion of the elbow.The hand consists of **carpals, metacarpals** and **phalanges.**

Carpals,the wrist bones, 8 in number are arranged in two rows of four each. The anterior surface of the wrist has tunnel-like appearance, due to the arrangement of carpals with the ligaments. This tunnel is termed as **carpal tunnel**.

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